TABLE 2. Bank-level trust bias and probability of sovereign exposure.

Outcome →	(1) Sovereign exposure	(2) Sovereign exposure	(3) Sovereign exposure	(4) Sovereign exposure	(5) Sovereign exposure
Bank-level trust bias	1.353***	1.757***	1.604***	1.630***	1.562***
	[0.110]	[0.329]	[0.301]	[0.300]	[0.310]
Bank-level branches			−0.090***	−0.153***	−0.163***
			[0.027]	[0.053]	[0.056]
Bank-level branches (squared)				0.014	0.016
				[0.011]	[0.012]
Bank × time FEs	Yes	Yes	Yes	Yes	Yes
Target-country × time FEs	Yes	Yes	Yes	Yes	No
Home-country × target-country FEs	No	Yes	Yes	Yes	No
Home-country × target-country × time FEs	No	No	No	No	Yes
Observations	23,760	23,760	23,760	23,760	21,615

Notes: The table summarizes the results of equation (1) estimated over the full sample period from 2010-Q1 to 2021-Q2. Dependent variable is *Sovereign Exposure*, defined as a dummy variable indicating any positive exposure of a bank toward a target country at a point in time reported in EBA and CEBS disclosures. *Bank-level Trust Bias* is computed for each bank–target country pair as the branch-weighted average (see equation (2)) of the residuals from a gravity model of trust (see equation (3)), where trust is defined as the portion of individuals in home country expressing "a lot of trust" toward target country, measured via Eurobarometer surveys. *Bank Branches* measures the number of bank branches (in thousands) that the bank owns in the target country. All branch-related information is from SNL Financial. For the detailed construction of the data, see Section 3. Robust standard errors are clustered at the bank level and reported in brackets. *Significant at 10%; **significant at 5%; ***significant at 1%.

5. Results

5.1. Baseline Results

Trust can be relevant for banks' investments in sovereign bonds for several reasons. Multiple countries make for multiple courts with uncertain jurisdiction. Governments enjoy a degree of sovereign immunity, casting doubt on the existence of judicial solutions. Such considerations may heighten reliance on trust as an alternative to legal contract enforcement. Such cultural stereotypes, defined in our context as how trustworthy residents of one nation view residents of another, may apply directly to sovereign bonds, since these are claims on governments representing specific nationalities. Table 2 reports estimates of equation (1) using our bank-level measure of trust biases. Column (1) includes a rich set of fixed effects but no other controls. Column (2) renders country-pair-specific controls redundant by adding *home-country* × *target-country* fixed effects. Columns (3)–(4) include controls for branch linkages between banks and target countries, both linear and nonlinear. Column (5) saturates the model with *home-country* × *target-country* × time fixed effects. This limits the comparison to banks headquartered in the same country with exposures to the same government at the same point in time. It thereby enables us to disentangle the effect of bank lending supply, our concern here, from demand-side factors in the countries to which banks lend.

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Estimates of the effect of bank-level trust bias are positive and statistically significant.[33] Point estimates grow larger as we add controls such as the number of bank branches and include country-pair fixed effects to capture other unobservables. The specification in column (5) flexibly controls for country-level unobservables by allowing them to vary over time.[34] Here a one standard deviation rise in bank-level trust bias is associated with an increase of 14% in the probability of investing in a target country. This is a large effect, accounting for one-third of the diversification gap (i.e., 42%) in banks' sovereign exposures and one-fourth of the unconditional mean of the dependent variable (i.e., 58%). We reach similar conclusions when employing the logarithmic value (in millions) of banks' sovereign exposures (instead of a binary indicator) as the dependent variable in panel A of Online Appendix Table C.3. Here, a one standard deviation rise in trust (based on column (5)) is associated with more than a 105% increase, more than doubling the volume of sovereign lending in a target country.[35]

Figure 2 plots the coefficients from separate estimates of equation (1) over subperiods. The positive relationship between bank-level trust bias and sovereign exposures, whether measured as discrete (panel A) or continuous (panel B), is significant and stable despite the changes in bank coverage. This observation is consistent with the intuition that cultural biases persist over time. It rules out the concern that our estimates are driven by Eurozone crises in the early part of our period or by changes in regulation over time.[36]

Our results remain intact when we exclude domestic (i.e., home country = target country) observations in Online Appendix Table C.5. This further shields our estimates from the influence of home bias (i.e., banks disproportionately holding the sovereign debt of their home countries). Although our baseline specification controls for home bias at the country level by including country-pair-specific fixed effects, one can imagine different degrees of home bias across different banks headquartered in the

33. The baseline estimate in column (1) is slightly larger than estimates using country-level trust measures reported in Online Appendix B. However, elasticities (in response to one std. dev. change) are approximately equal (≈12%) in both cases.

34. Note that including *home-country* × *target-country* × *time* fixed effects shields our estimates from the possibility of home bias (i.e., banks generally holding higher sovereign debt of their home countries) even when such bias is heterogenous across countries and varying over time. Our estimates remain significant at conventional levels with double clustering at country-pair and time levels (see Online Appendix Table C.1) or at country-pair and bank levels (see Online Appendix Table C.2).

35. Panel B of Online Appendix Table C.3 presents the intensive margin estimates where exposures with zero value are excluded and reports qualitatively similar results.

36. The reduction in the size of the coefficients in the period 2016 to 2018 (shaded in Figure 2) confirms that loss of granularity (due to changing reporting requirements during this period) in banks' sovereign exposures makes it more difficult to identify the effect of trust. EBA directly used regulatory FINREP reports during this period, which led to some banks not disclosing the country breakdown of their sovereign exposures at all or reducing the granularity in these exposures (i.e., categorizing exposures below a certain threshold under the name "other countries"). In line with Figure 2, our point estimate in column (5) of Table 2 becomes approximately 12% larger when we drop FINREP disclosure dates from our sample (see Online Appendix Table C.4).

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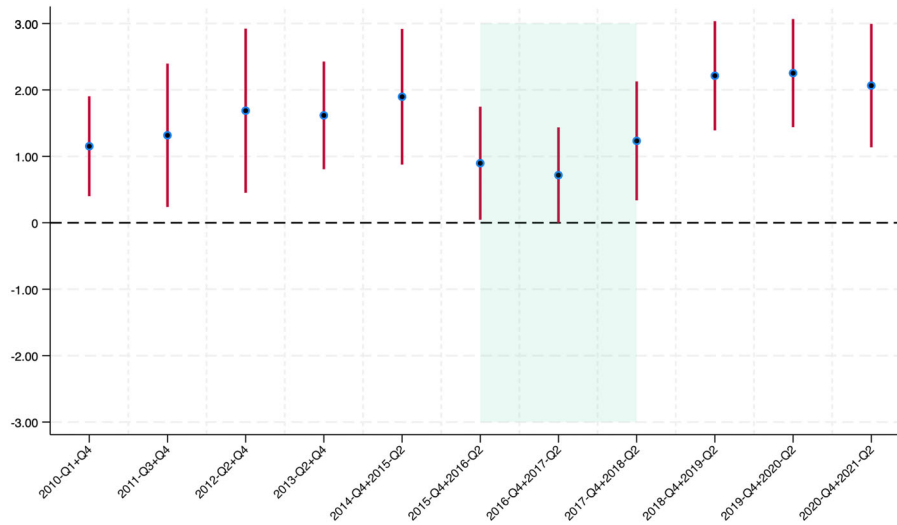
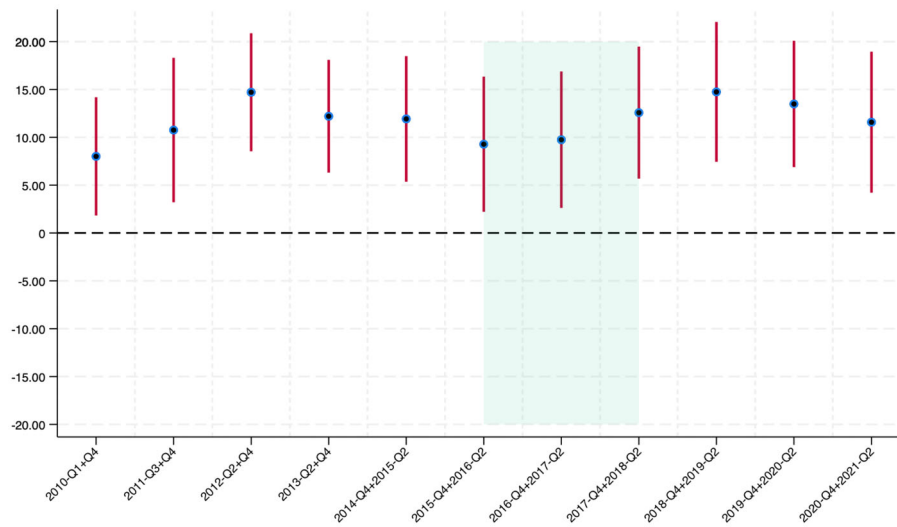


(a) Dependent variable: Sovereign exposure (dummy)



(b) Dependent variable: Sovereign exposure (log nominal)

FIGURE 2. Baseline estimates over subsample periods. This figure shows estimates for the coefficient of bank-level trust bias separately for eleven distinct subsample periods. Dependent variables are the probability of sovereign exposure (upper panel) and log nominal sovereign exposures—in millions—(lower panel). Shaded areas indicate subperiods during which EBA reported sovereign exposures based on regulatory FINREP data that restrict the level of granularity disclosed in banks' sovereign debt portfolios. The specification is column (5) of Table 2. Only the estimated coefficient on *Bank-level Trust Bias* is plotted. Confidence intervals are at 90% significance level. Source: EBA, CEBS, Eurobarometer, and SNL Financial.

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same country. However, when we exclude investments in the bonds of the home country, the association between cultural biases and foreign government exposures is, if anything larger than indicated by the baseline estimates in Table 2.

Regulators may prevent banks from investing in countries in which they have little trust. They may discourage banks from investing in foreign government bonds as a way of encouraging them to invest in their own country's government bonds. However, we obtain similar results, as shown in Online Appendix Table C.6, when limiting the sample to banks overseen by the EU's SSM. Focusing on banks supervised by the SSM thus rules out the alternative hypothesis that we are picking up the cultural stereotypes (or other idiosyncrasies) of local bank supervisors as opposed to bankers.[37]

Exchange rate risk could affect the decision to invest and thus the extensive margin of banks' sovereign exposures. Eurozone-headquartered banks may be inclined to invest in the bonds of Eurozone governments while refusing to invest in the local-currency bonds of other countries subject to exchange risk. In Online Appendix Table C.7, we therefore include only banks headquartered in the Eurozone and target countries of investment that are members of the Eurozone. The results carry over.

Banks situated in the Eurozone's crisis countries—Greece, Italy, Ireland, Portugal, and Spain—were subject to financial problems at the beginning of our sample period, which could have affected banks' investment decisions. While Figure 2 speaks against this concern by showing that our results extend beyond the crisis period, we can directly exclude both banks headquartered in these countries and their governments as potential targets for cross-border investment. There is again little change in our results (Online Appendix Table C.8). This further rules out the possibility that the patterns we observe are driven by the large spreads or relatively generous treatment under supervisory stress tests of the bonds of these countries.[38]

Our empirical setting with banks' branch networks and a weighted computation of trust biases could conceivably be conducive to mechanically generating the results in Table 2. Online Appendix Table C.12 therefore presents two placebo tests in which we randomly distribute observed branch networks either across banks located in the same home country (panel A) or across all banks in our sample (panel B). Both tests confirm that the previous findings are not an artifact of our empirical setting.

37. The SSM, housed in the ECB in Frankfurt, supervises more than 100 of the largest banks in Europe; the same supervisors apply the same rules and scrutiny to all of them. Since SSM started its operations in 2014, there has been limited time variation in terms of the number and identity of systemically significant banks that it supervises. Our results in Appendix Table C.6 take this time variation into account, although the results are very similar if we focus only on the initial set of banks that came under the supervision of SSM in 2014.

38. In Online Appendix Table C.9, we substitute trust bias (as in Online Appendix Table A.5) with the trust measure in levels (as in Online Appendix Table A.4), considering the simple proportion of people in a country with "a lot of trust" toward another country aggregated at the bank level. Online Appendix Table C.10 substitutes the *graded* cultural trust bias proxy employed in Guiso et al. (2009), which uses the full variation in survey respondents' answers ranging from 1 (i.e., "no trust at all") to 4 (i.e., "lot of trust"). In Online Appendix Table C.11, we use the *graded* proxy in levels without computing the residuals as in equation (3). Results carry over.

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